

Creating land surface datasets for E3SM - Internal

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Overview

The following steps describe the method for generating a surface dataset for use in E3SM based on LUH2 data. This process involves converting the LUH2 data into an LUH1 format, generating associated land cover changes using the Land Use Translator (LUT), and then using these land cover changes with mksurfddata to generate the actual surface dataset for E3SM. All existing CMIP6 scenarios (through 2100) have already been converted from LUH2 to LUH1 format. The required (and already generated) files are currently on NERSC, COMPY, and BLUES (including the public html data server: [Index of /public/e3sm/inputdata](#)). Please check in `.../inputdata/lnd/clm2/surfddata_map` to find out if the most recent version of your desired land surface files may already exist. Also check `.../inputdata/lnd/clm2/rawdata` to find out if current LUT output files exist for a particular scenario. The current LUT version was finalized in early July 2020 as part of iESMv2, and is named `updateannuallanduse_v2.c`. LUT output files and land surface files dated 8 July or later should be current, but if in doubt use the file with the most recent date, or ask the file owner if they know whether or not the file was created with the new LUT.

Note that all NetCDF files should be in version 3 format.

The new baseline historical land surface files for ne30np4 resolution have been created with the current LUT version, as well as two future scenarios (SSP5 RCP8.5 and SSP2 RCP4.5), and are on NERSC, COMPY, and BLUES in `.../inputdata/lnd/clm2/surfddata_map`. Transient-consistent static surface files for 1850, 2000, and 2015 are also present for ne30np4. Additional files have been and will be added as created, and all available files are listed below. Note that any single-year surfddata files below without a scenario tag are associated with the 'HIST' landuse.timeseries files (these were distributed before full naming conventions were adopted). All other surfddata files should have scenario tags that match the corresponding landuse.timeseries files.

The following new land surface files are available on NERSC, COMPY, and BLUES:

- landuse.timeseries_ne30np4_HIST_simyr1850-2015_200728.nc
- landuse.timeseries_ne30np4_SSP2_RCP45_simyr2015-2100_200728.nc
- landuse.timeseries_ne30np4_SSP5_RCP85_simyr2015-2100_200728.nc
- surfdata_ne30np4_simyr1850_c200729.nc
- surfdata_ne30np4_simyr2000_c200729.nc
- surfdata_ne30np4_simyr2015_c200812.nc
- landuse.timeseries_0.5x0.5_HIST_simyr1850-2015_c200924.nc
- landuse.timeseries_0.5x0.5_SSP1_RCP26_simyr2015-2100_c220317.nc
- landuse.timeseries_0.5x0.5_SSP5_RCP85_simyr2015-2100_c220318.nc
- surfdata_0.5x0.5_simyr1850_c200924.nc
- surfdata_0.5x0.5_HIST_simyr2015_c220318.nc
- surfdata_0.5x0.5_SSP1_RCP26_simyr2100_c220317.nc
- surfdata_0.5x0.5_SSP5_RCP85_simyr2100_c220318.nc
- landuse.timeseries_0.9x1.25_HIST_simyr1850-2015_c201021.nc
- landuse.timeseries_0.9x1.25_SSP5_RCP85_simyr2015-2100_c201021.nc
- surfdata_0.9x1.25_HIST_simyr1850_c201021.nc
- surfdata_0.9x1.25_HIST_simyr2000_c201021.nc
- surfdata_0.9x1.25_HIST_simyr2015_c201021.nc

The data units vary slightly between the initial inputs and the processed files. The LUT input files, which are in LUH1 format, have units of fraction of grid cell. The LUT processes these data into percent of vegetated land unit for PFTs, and into fraction of vegetated land unit for harvest. Note that the LUH urban data are not used, and that the harvest data are normalized based on the sum of LUH primary and secondary land, which represents vegetated area (and is the basis for calculating the area harvested from the biomass harvested). Separate urban data are used by mksurfdata to reduce the vegetated land unit to make room for the urban land unit. The E3SM land surface files also have units of percent of vegetated land unit for PFTs, and fraction of vegetated land unit for harvest. Note that E3SM currently sums the five harvest type fractions (because primary and secondary land are not currently distinguished in E3SM and wood harvest comes from trees) and applies this total to only the tree PFTs to obtain harvested wood carbon.

Another important thing to note is the time that these data represent in relation to their labeled year. Both the PFT and the harvest data are stored by year in the LUT output and land surface files, but they represent two respective, different time periods. The stored PFT values represent a snapshot at the beginning of the labeled year, and throughout the model year values are interpolated between two consecutive years. For example, during model year 2015, the file PFT values for years 2015 and 2016 are used to interpolate the PFT conversion for a given time step. On the other hand, the file harvest data represent an annual rate for the year prior to the labeled year, and are adjusted in the model based on the desired harvest frequency during the year. For example, the harvest data for model year 2015 are stored and labeled as 2016 in the files, and are applied in 2015. This is a legacy design in data generation and use that provides the next PFT and harvest data together with the same time label at the beginning of the model year.

These instructions are updated from a previous version at [Provenance for CMIP6\(Future\) Land Use for E3SM](#), as the current LUT has been updated in July 2020.

More complete mksurldata instructions are at [CLM4.5 dataset for year 1850 at ne120np4 resolution](#) if you need to generate a new E3SM grid resolution.

The following procedure for generating land surface datasets is straightforward, but includes steps that may be unfamiliar to some folks. If you have any questions please contact Alan Di Vittorio at avdivittorio@lbl.gov. In some cases it may be more efficient for Alan to generate a dataset than for someone else to learn the process.

Converting LUH2 datasets to LUH1 format

LUH2 data (CMIP6) include historical data at three levels of land use, eight future scenarios, three future extensions beyond 2100, and added tree cover for future scenarios (available from <https://luh.umd.edu/data.shtml>). ELM currently supports land data only in LUH1 format, which is at 0.5 degree spatial resolution with five land use types and five harvest categories, and with units of fraction of grid cell. The conversion of LUH2 data to LUH1 format is described here, and all CMIP6 scenarios have already been converted to LUH1 format. Converted data can be found at NERSC, COMPY, and BLUES in the `.../inputdata/ln/clm2/rawdata/LUT_input_files_current/directory` and are processed further as described in the following sections.

For the spatial aggregation, quarter-degree land use fractions within each half-degree grid cell are averaged to get the fractions of land use types in each half-degree grid cell. This is consistent with the method used for previous E3SM historical land use datasets (although for v3 a more accurate aggregation method that directly calculates the actual half-degree fractions should be used; based on limited tests the differences between averaging and direct calculation

in some grid cells can be over 2%). This difference is well within uncertainty and as such is acceptable.

For the aggregation of land use types into the five LUH1 categories, the LUH2 types c3ann, c4ann, c3per, c4per, and c3nfx are combined into a single cropland

category (GCROP), the LUH2 types pastr and range are combined into a single pasture category (GPAST), the LUH2 types primf and primn are combined into a

primary vegetation category (GOTHR), the LUH2 types secdf and secdn are combined into a secondary vegetation category (GSECD), and the LUH2 type urban is simply relabeled as the LUH1 type GURBN. The harvest categories are the same in LUH2 and LUH1.

Using the Land Use Translator (LUT) to convert LUH1 format data to ELM plant functional types (PFTs) and harvest fractions

The Land Use Translator (LUT) is software that translates the land use data from LUH1 into land cover data, in terms of the 16 PFTs used by CLM, and normalizes the harvest data to fraction of vegetated land unit. The LUT outputs are at 0.5 degree spatial resolution and have units of either percent (for PFTs) or fraction (for harvest) of vegetated land unit.

The LUT was originally written by Peter Lawrence at NCAR, was subsequently modified for use in the iESM by Alan Di Vittorio at LBL, has undergone more modification for use with E3SM by Ritvik Sahajpal and Louise Chini at UMD, and has been modified again by Alan Di Vittorio to arrive at its current version for E3SM v2. This version is identical for both standalone and iESM creation of land surface datasets, and produces identical results across machines and compiler optimizations. The land conversion preferences in the LUT are currently set to best match qualitative examination of SSP5 RCP8.5 scenario forest area changes while potentially reducing undesired behavior in other scenarios (i.e., proportional addition and removal of PFTs). This version also keeps track of existing pasture when converting land each year.

The current version of the LUT code is now available in the E3SM GitHub repository in:

[E3SM/components/elm/tools/clm4_5/land_use_translator/](https://github.com/E3SM-Project/E3SM/tree/master/components/elm/tools/clm4_5/land_use_translator/).

The LUT currently requires all land-use data to be converted into LUH1 format (0.5 degree spatial resolution, annual time-steps, and using the following 4 or 5 land use types: cropland, pasture, primary vegetation, secondary vegetation, and optionally urban), as detailed in the section above. Note that the urban land-use data from LUH1 is not actually used as an input to the LUT, so it is indirectly converted into one of the natural vegetation PFTs in LUT. This is because separate urban area data are used by mksurfdata.

The 16 PFTs output by the LUT, along with their numeric index and description, are:

- 0 BPFT, Bare ground/not vegetated
- 1 NEMPFT, Needleleaf evergreen temperate tree
- 2 NEBPFT, Needleleaf evergreen boreal tree
- 3 NDBPFT, Needleleaf deciduous boreal tree
- 4 BETPFT, Broadleaf evergreen tropical tree
- 5 BEMPFT, Broadleaf evergreen temperate tree
- 6 BDTPFT, Broadleaf deciduous tropical tree
- 7 BDMPFT, Broadleaf deciduous temperate tree
- 8 BDBPFT, Broadleaf deciduous boreal tree
- 9 SEMPFT, Broadleaf evergreen temperate shrub
- 10 SDMPFT, Broadleaf deciduous temperate shrub
- 11 SDBPFT, Broadleaf deciduous boreal shrub
- 12 GA3PFT, C3 arctic grass
- 13 GC3PFT, C3 non-arctic grass
- 14 GC4PFT, C4 grass
- 15 CPFT, Cropland

Note that there is space in the PFT array for a 17th PFT that is not used.

~~Running the LUT~~

The default mode for the LUT code is a standalone version that creates inputs to mksurfddata from LUH1 format data. First load the netcdf module and run the Makefile, and then pass the name of the land use states file and the directory of input files (and optionally the output path as argument three) to the LUT as command-line arguments. The harvest file must be named identically to the states file but with "_harvest" before the ".nc" because it is inferred by the code based in the states file name. The historical file names MUST CONTAIN the string "HIST" (in caps), and future file names MUST NOT CONTAIN "HIST" because this is how the years to process are determined.

For example (on NERSC), to generate the SSP5 RCP8.5 files:

```
2 make
3 ./land_use_translator LUH2_SSP5_RCP85_LUH1f_c07182019.nc \
4 /global/cfs/cdirs/e3sm/inputdata/ln2/clm2/rawdata/LUT_input_files_current/
```

There is also an optional third argument that is the path to the output files, which will otherwise go into a new directory (./output) by default.

On COMPY one has to first load a compiler and then netcdf before running 'make.' For example:

```
1 module load intel
2 module load netcdf
```

Also, on COMPY the input files are in the corresponding /compyfs/inputdata/ln2/clm2/rawdata/LUT_input_files_current/ directory.

The following CMIP6 LUH2 data are currently available in LUH1 format. The state file names are:

- LUH2_HIST_LUH1f_c07182019.nc (1850-2015)
- LUH2_HIST_lo_LUH1f_c08292020.nc (1850-2015)
- LUH2_HIST_hi_LUH1f_c08292020.nc (1850-2015)
- LUH2_SSP1_RCP19_LUH1f_c08292020.nc (2015-2100)
- LUH2_SSP1_RCP26_LUH1f_c08292020.nc (2015-2100)
- LUH2_SSP2_RCP45_LUH1f_c08292020.nc (2015-2100)
- LUH2_SSP3_RCP70_LUH1f_c08292020.nc (2015-2100)
- LUH2_SSP4_RCP34_LUH1f_c08292020.nc (2015-2100)
- LUH2_SSP4_RCP60_LUH1f_c08292020.nc (2015-2100)
- LUH2_SSP5_RCP34_LUH1f_c08292020.nc (2015-2100)
- LUH2_SSP5_RCP85_LUH1f_c07182019.nc (2015-2100)

Note that the initial years for the time series (1850, 2015) are inferred by the file name as described above. This assumes that the initial 1850 files already exist (which they do for the baseline historical data).

To create 1850 initial files for the LUT and for mksurfddata, use '1850' as the first argument in place of a file name. This runs a single year historical run that sets up 1850 initial files with the correct year. The output 1850 file should be added to the directory of historical file LUT outputs

for use by mksurfddata in generating an initial 1850 land surface file (which is useful for spinning up to initial conditions).

If you want a different historical state input file you should create new 1850 initial files to go along with it. This also applies if you change this code (although usually only the mcrop input file needs to be updated in the case of only **changing code**).

To do so, first update the default historical state file name in the code, compile, and then enter "1850" in place of the file name argument (otherwise the default files will be used). **This step isn't necessary if only changing this code.**

Search for the following to find the line to update:

- `sprintf(usesfname, "LUH2_HIST_LUH1f_c07182019.nc");`

Second, create new 1850 initial files by replacing the data in the original files with the 1850 data from the two new dynamic LUT files. The original input files are currently named:

- `iESM_Ref_CropPast1850_c10142019.nc`
- `surfddata_360x720_mcrop1850_c07082020.nc`

and the dynamic files have either "Dyn" (land use data) or "dyn" (PFT data) in the name.

Third, update the new 1850 initial file names in the code below (assuming you changed the creation data), compile, and then run using the same new historical file name as in the first step.

Search for the following to find the lines to update:

- `const char initial_hist_dyn_luh_file[] = "iESM_Ref_CropPast1850_c10142019.nc";`
- `const char initial_hist_dyn_pft_file[] = "surfddata_360x720_mcrop1850_c07082020.nc";`

The LUT outputs one file for each year that includes the PFT percent of vegetated land unit and the harvest fraction of vegetated land unit at half-degree resolution, and these files are input to mksurfddata as described below. The two additional output files are the dynamic files used by the LUT in creating the annual output files (and which are the source of 1850 initial file data when running the LUT with the '1850' option).



Generating an ELM surface dataset with mksurfddata

After generating the annual land cover (in PFTs) using the LUT, the next step is to generate a surface dataset that is based on these data. This surface dataset places the PFT changes onto the land surface grid used by E3SM and also includes data on other surface properties such as soils, slopes, glaciers, wetlands, etc.

The mksurfddata software is used to generate the surface dataset. Before you can run mksurfddata, you will first need to set up the input files (from the LUT) and clone the E3SM code repository from GitHub (the instructions in the Development Getting Started Guide are a useful resource for cloning E3SM).

You can follow these steps to generate the surface dataset, assuming that you have cloned E3SM into your \$SCRATCH directory:

1. Set up the input files:

All input files (which are the LUT annual outputs) for a given time series need to be listed in a single text file with one filename per line and the year of the file starting at exactly character 197 (the code expects the first 195 characters to be the filename with full path, then a space, then the year). The name of this text file will be the mksrf_fdynuse namelist value for mksurfddata (see step 5 below). It is recommended that files for years 1850-2015 are in a historical directory, years 2016-2100 in a future directory, and that the each text file is stored in the directory with the files it points to. Note, that the text file for future scenarios should point to the 2015 file in the historical directory to have a clear record of the starting point. For example, the text file for generating an SSP5 RCP8.5 dataset is:

```
.../inputdata/lnd/clm2/rawdata/LUT_LUH2_SSP5_RCP85_LUH1f_07082020/LUT_LUH2_SSP5_RCP85_LUH1f_list.txt
```

This text file includes the complete path to the files, which is both machine and user dependent and so precludes a standard for automatically creating this file. It is recommended to copy an existing file (such as the example above) and modify it appropriately.

2. Load the relevant modules and configure the environment (e.g, on NERSC):

```
1 module load cray-netcdf
2 module load cray-hdf5
3 export LIB_NETCDF=$NETCDF_DIR/lib
4 export INC_NETCDF=$NETCDF_DIR/include
5 export USER_FC=ifort
6 export USER_CC=icc
7 export USER_LDFLAGS="-L$NETCDF_DIR/lib -lnetcdf -lnetcdff -lnetcdf_intel"
8 export USER_LDFLAGS=$USER_LDFLAGS" -L$HDF5_DIR/lib -lhdf5 -lhdf5_fortran -lhdf5_cpp -
9 lhdf5_fortran_intel -lhdf5_hl_intel -lhdf5hl_fortran_intel"
10
```

On COMPY the environment requires a slightly different set of commands to configure:

```
1 module load intel
```



```

2 module load netcdf
3 module load hdf5
4 export LIB_NETCDF=$NETCDF_ROOT/lib
5 export INC_NETCDF=$NETCDF_ROOT/include
6 export USER_FC=ifort
7 export USER_CC=icc
8 export USER_LDFLAGS="-L$NETCDF_ROOT/lib -lnetcdf -lnetcdff "
9 export USER_LDFLAGS=$USER_LDFLAGS" -L$HDF5_ROOT/lib -lhdf5 -lhdf5_fortran -lhdf5_cpp -
  lhdf5_hl -lhdf5hl_fortran "

```

3. Compile the mksurfddata code:

```

1 cd <e3sm-dir>/components/elm/tools/clm4_5/mksurfddata_map/src/
2 gmake clean
3 gmake

```

4. Run the mksurfddata.pl script in “debug” mode to generate the namelist:

Select the desired land resolution and which year to generate a single surface file, such as the initial or final year surface data. The time series will be based on the mksrf_fdynuse namelist value (which you will set manually), and only one year of the -years argument will be processed (-years denotes which file names to select for the fvegtype and fsurdat namelist options for the single-year surface file to be created; see step 5 below). For historical data omit the -rcp argument. **Two separate historical mksurfddata runs (steps 4-6) are needed to create both an1850 initial surface file for spinup and historical simulations and a 2015 initial surface file for future simulations.** The -rcp argument is used to select a file for -years and has been updated for land surface file creation to include 8 possible SSP-RCP scenarios: “1-1.9”, “1-2.6”, “4-3.4”, “5-3.4OS”, “2-4.5”, “4-6.0”, “3-7.0”, “5-8.5”. **Note that not all scenarios have LUT files available yet for use by mksurfddata, and that these -rcp options are currently unique to the -dincl flag for land surface file creation.** The namelist_defaults_clm4_5_tools.xml file (in E3SM/components/elm/bld/namelist_files/) does include -year 2015 defaults for mksrf_fvegtype so that a namelist file can be created for each scenario without having to modify this defaults file, but mksrf_fvegtype, fsurdat, and fsurlog will have to be modified in the namelist to generate the desired single-year output file (see step 5 below). As LUT files are created for additional scenarios, namelist_defaults_clm4_5_tools.xml can be updated to include all the available files for the namelist. The -d flag denotes a debug run that only creates the namelist. The namelist file is named “namelist” and is created in same directory as the mksurfddata.pl script. For example, to generate the namelist for an SSP5 RCP8.5 dataset at ne30np4 resolution and its final surface data file, enter on the command line (e.g., NERSC):

```

1 cd <e3sm-dir>/components/elm/tools/clm4_5/mksurfddata_map/
2
3 ./mksurfddata.pl -res ne30np4 \
4 -years 2100 -rcp 5-8.5 -d \

```

```
5 -dinlc /global/cfs/cdirs/e3sm/inputdata \  
6 -usr_mapdir /global/cfs/cdirs/e3sm/inputdata/ln/clm2/mappingdata/maps/ne30np4
```

On COMPY, the inputdata are in a different location:

```
1 cd <e3sm-dir>/components/elm/tools/clm4_5/mksurfddata_map/  
2  
3 ./mksurfddata.pl -res ne30np4 \  
4 -years 2100 -rcp 5-8.5 -d \  
5 -dinlc /compyfs/inputdata \  
6 -usr_mapdir /compyfs/inputdata/ln/clm2/mappingdata/maps/ne30np4
```

Note that this step will fail if the required mapping files are not in `../inputdata/ln/clm2/mappingdata/maps/<res>`, or if the definitions of these mapping files are incorrect in `namelist_defaults.xml` (search for the “map” tag and the “to_hgrid” field that matches your desired resolution, e.g., `to_hgrid=“ne30np4”`). You should also check in `namelist_defaults_clm4_5_tools.xml` to make sure that there is a “scripgriddata” tag entry with an “hgrid” field that matches the desired resolution.

5. Modify mksurfddata namelist entries:

Five namelist options need to be filled or modified in the “namelist” file:

- **mksrf_fvegtyp**: The source PFT file for the single-year surface data file to be created (fsurdat, which can be an initial file for constant or transient land surface simulations). This source file is an LUT output file. This entry may not be correct by default if `namelist_defaults_clm4_5_tools.xml` has not been updated to include the desired output year (see step 4 above).
- **mksrf_fdynuse**: The text file listing all of the LUT output files to be used for creating the land surface file (see step 1 above).
- **fdyndat**: The name of the transient land surface file to be created. The recommended naming format is (replace brackets and bracketed text):
`landuse.timeseries_<res>_<scenario>_simyr<yyyy>-<yyyy>_c<yymmdd>.nc`
- **fsurdat**: The name of the single-year land surface file as determined by the -years argument to mksurfddata.pl (step 4 above) and mksrf_fvegtyp source file. Make sure the years match between fsurdat and mksrf_fvegtyp namelist entries. For future simulations, the default name will not include the SSP and RCP numbers (e.g., 5-8.5), and may not have the correct year if `namelist_defaults_clm4_5_tools.xml` has not been updated to include the desired output year

(see step 4 above). It is recommended to update this so as to not cause confusion among scenarios. The recommended naming format

is: `surfdata_<res>_<scenario>_simyr<yyyy>_c<yymmdd>.nc`

- `fsurlog`: The name of the log file for `fsurdat`, so make sure it matches, but with a `.log` extension in place of `.nc`.

For example, to generate the SSP5 RCP8.5 transient land surface file and year 2100 land surface file (see step 4) set the following namelist entries to:

- `mksrf_fvegtyp =`
`'.../inputdata/Ind/clm2/rawdata/LUT_LUH2_SSP5_RCP85_LUH1f_07082020/LUT_LUH2_SSP5_RCP85_2100_c07082020.nc'`
- `mksrf_fdynuse =`
`'.../inputdata/Ind/clm2/rawdata/LUT_LUH2_SSP5_RCP85_LUH1f_07082020/LUT_LUH2_SSP5_RCP85_LUH1f_list.txt'`
- `fdyndat = 'landuse.timeseries_ne30np4_SSP5_RCP85_simyr2015-2100_c200812.nc'`
- `fsurdat = 'surfdata_ne30np4_SSP5_RCP85_simyr2100_c200812.nc'`
- `fsurlog = 'surfdata_ne30np4_SSP5_RCP85_simyr2100_c200812.log'`

Note the relationship between the `mksurfdata.pl` run (step 4) and the required namelist entries (this step). In general, if `-years` or `-rcp` in step 4 do not provide the correct namelist entries, this can be remedied by modifying the five namelist items above. For example, the `$SCRATCH/E3SM/components/elm/bld/namelist_files/namelist_defaults_clm4_5_tools.xml` file does not yet contain `mksrf_fvegtyp` entries for all scenarios and years (because all the files are not yet available). It does, however, contain an entry for 2015 for the baseline historical and all 8 future scenarios so that a namelist file can be created, and then modified to point to the desired files as available.

6. Create the land surface dataset:

Running `mksurfdata_map` will now generate a transient land surface file, a single-year land surface file, and a log file for the single-year land surface file. For example, to create the SSP5 RCP8.5 files based on the steps above, enter on the command line (assuming the current directory is `$SCRATCH/E3SM/components/elm/tools/clm4_5/mksurfdata_map/`):

```
>> ./mksurfddata_map < namelist
```

If a different single-year land surface file is desired, modify mksrf_fvegtyp, fsurdat, and fsurlog in the namelist file (see step 5) and re-run mksurfddata_map. The transient land surface file will be overwritten, so it may be prudent to modify fdyndat also, especially if the date is different.

7. Copy the LUT and land surface files to the inputdata directory:

This enables you and others to use the new LUT and land surface files. It also reduces duplication of land surface files if others can find an existing file in the inputdata directory. If a current, single-year land surface file already exists in inputdata that matches the one you just created, you should copy only the transient file. In other words, please try to not create and copy the same data files with different names or creation dates. Just copying the files to `../inputdata/ln/clm2/surfddata_map/` should be sufficient on COMPY, but on NERSC you must change the group of the copied file to e3sm for others to use it (`chgrp e3sm <filename>`).

The LUT files for a given scenario, including the text file, should go into a specific directory within `/inputdata/ln/clm2/rawdata/` (see step 1 above for an example). The recommended naming convention for future scenario directories is `LUT_LUH2_SSP<#>_RCP<##>_LUH1f_<mmddyyyy>`, and for historical scenarios is `LUT_LUH2_HIST_<tag>_LUH1f_<mmddyyyy>`.

If the files you have created are correct and final, please notify Alan Di Vittorio so that they can be transferred to the main data server on BLUES.

E3SM namelist options for running a transient simulation

Three additional namelist variables must be set to enable a full transient simulation (following the example above):

- `do_harvest = .true.`
- `do_transient_pfts = .true.`
- `flanduse_timeseries = '<path to inputdata>/inputdata/ln/clm2/surfddata_map/landuse.timeseries_ne30np4_SSP5_RCP85_simyr 2015-2100_c200812.nc'`

The single-year land surface file used for the initial land configuration (fsurdat) may not have the same PFT distribution as the corresponding year the in transient land surface file (flanduse_timeseries). Ideally these would match, but for some experiments matching PFTs between these files may not be feasible or even desirable.

When the transient PFTs do not match the fsurdat PFTs, set the namelist option:

- `check_dynpft_consistency = .false.`

In some cases the initial data file (finidat) may not match the single-year land surface file (fsurdat) in name and/or date, causing a run-time error. In such cases where these inconsistencies are known and deemed appropriate (e.g., a filename change or initial conditions are purposefully from a control simulation with a different land surface) set the following namelist options:

- `check_finidat_fsurdat_consistency = .false.`
- `check_finidat_year_consistency = .false.`

Please make sure that these inconsistencies are scientifically valid before allowing them.